

Bio efficacy of Indigenous isolates of *Beauveria bassiana* against different Stages of *Bactrocera zonata* (Guava Fruit fly)

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Abstract

Background Pakistan is the world's leading producer of guava, with 0.0585 million hectares under cultivation, yielding a total production of 0.56 million tons guava annually. However, various biotic and abiotic constraints limit global guava yields. Among the biotic factors, the fruit fly, *Bactrocera zonata* poses a significant challenge in Pakistan. The indiscriminate use of chemical insecticides to control these pests has caused significant environmental and regulatory concerns, highlighting the need for ecologically safe and environmentally friendly alternatives. In this context, native isolates of entomopathogenic fungus, *Beauveria bassiana* have been isolated and their efficacy was evaluated against different stages of *Bactrocera zonata*.

Results Susceptibility of the fruit fly varied significantly with the spore concentration. The indigenous isolate, FMB-BB-FF 04, demonstrated high virulence against the 3rd instar larvae, pupa and adult causing up to 98% mortality at 14 days after treatment (DAT) at the highest spore concentration tested (1×10^8 conidia/mL).

Furthermore, horizontal transmission of fruit fly adult was tested achieving highest mortality rates when both male and female were inoculated. While comparatively higher mortality was recorded in treatments with inoculated females than treatments with non-inoculated females. FMB-BB-FF 02 and FMB-BB-FF 04 exhibited high mortality, suggesting its potential as a bio-pesticide in fruit fly management programs.

Conclusions Based on virulence, and efficacy, FMB-BB-FF 04, showed significant potential as a biocontrol agent for managing guava fruit fly. The present study identified an effective native isolate and demonstrated its in vitro efficacy. Thus, it could be integrated into fruit fly management programs in guava cultivation as an efficient biological control agent.

Background

Bactrocera zonata is a significant global pest which is responsible for economic losses. It is one of the most dominant, destructive, and prolific specie which capable of causing significant losses in guava cultivation (Murtaza et al. 2021). In different countries of the world, 10–80% fruit losses cause by *B. zonata* have been documented (Ahmad and Begum 2017). In Pakistan, it is responsible for up to 89.50% losses (Murtaza et al. 2021). It is native to Asia and is found in several ASEAN countries, as well as in Thailand, Bangladesh, India, Sri Lanka, Egypt, and Mauritius (El-Minshawy et al. 2018). Its population peaks during temperate seasons, increasing the risk of infestation (Murtaza et al. 2022). To manage harmful fruit fly populations, growers primarily depend on residual insecticides applied directly to foliage or used as bait sprays to target adult flies, as well as soil treatments beneath tree canopies to eliminate larvae and pupae. Besides chemical control, various IPM strategies are employed, including the male annihilation technique, use of parasitoids, cultural practices like orchard sanitation, and SITs. (Mangan 2014). However, the injudicious use of insecticides has raised concerns about reduced effectiveness due to development of resistance, human health hazards and negative impacts on non-targeted individuals

(Sabbahi and Hock 2024). This highlights the importance of expanding biological methods to support sustainable pest management (Usman et al. 2021). Among such alternatives, the use of entomopathogens has been investigated as a biological strategy to manage various insect pests, especially fruit flies (Gulzar 2021). In recent years, entomopathogenic fungi (EPFs) have gained significant attention as significant microbial biocontrol agents (Sutanto et al. 2023). These are environmentally friendly, biodegradable, having specific mode of action, and capable of overcoming insect resistance, especially with advancements in mass production techniques (Sabbahi et al. 2022). Research has also investigated the effectiveness of EPFs against various fruit fly species, including the Mediterranean fruit fly (*Ceratitis capitata*), the Mexican fruit fly (*Anastrepha ludens*), the European cherry fruit fly (*Rhagoletis cerasi*), the western cherry fruit fly (*R. indifferens*), *Bactrocera oleae*, and *R. pomonella*. Nevertheless, there has been limited research on the use of EPF isolates against *Bactrocera zonata*, *B. cucurbitae* and *B. dorsalis* (Usman et al. 2021). Although EPFs have demonstrated virulence against fruit flies, considerable variability exists among different isolates belonging to EPF species. This variability could be due to many factors like genetic differences, co-evolved with specific hosts in distinct geographical regions, and the original host species where fungi were isolated (Quesada-Moraga et al. 2012). Screening locally sourced EPF strains to identify promising isolates for field application could offer advantages in terms of cost-effectiveness and practical use by growers. However, studies investigating the efficacy of native EPF strains against *B. zonata* and *B. dorsalis* remain scarce in literature. In Pakistan, although research on biological insect control remains limited, local isolates of entomopathogenic fungi have been identified and evaluated for their pathogenicity against whitefly (Wakil et al. 2018). While entomopathogenic fungi (EPFs) have proven effective against tephritid fruit flies, significant variation exists in the virulence of different EPF isolates and species. This variability can stem from differences in genetic makeup and the adaptation of EPFs to hosts in specific geographic regions (Quesada-Moraga et al. 2012). Owing to their inherent genetic diversity and local adaptability, native isolates of *Beauveria bassiana* contribute significantly to the development of effective biocontrol agents. Numerous studies have shown that native isolates often perform better than non-native strains (Riaz et al. 2024). Native EPF isolates also exhibit a compatible host-pathogen relationship and significant efficacy against local pest species (Becher et al. 2018). These isolates typically pose fewer environmental risks, are cost-effective to mass-produce, and integrate more harmoniously with local ecosystems (Yee et al. 2009). Moreover, they offer insights into unique pest management mechanisms and help fill research gaps regarding local fungal ecology in underrepresented regions. Prioritizing native isolates could pave the way for sustainable pest control solutions. Hajek and Delalibera (2010) emphasized the importance of exploring native fungi to maintain ecological balance and develop region-specific pest management strategies. Additionally, Rangel et al. (2014) showed that native strains often possess unique metabolic profiles, which could prove valuable for innovative pest management approaches. Currently, there is limited literature on the potential of indigenous entomopathogenic fungi (EPF) against *Bactrocera zonata*. Many earlier studies have focused primarily on commercial strains, overlooking the genetic diversity and potential benefits of native isolates. Exploring local isolates can reveal novel genetic or metabolic traits that may help overcome challenges such as reduced efficacy under adverse climatic conditions. Such discoveries could address gaps in previous research, which

sometimes failed to capture the full potential of these fungi. Given the increasing demand for sustainable pest management strategies, the present study aims to evaluate the pathogenicity of indigenous isolates of *Beauveria bassiana* against different developmental stages of *B. zonata* in laboratory conditions. The study also investigated the horizontal transmission of different indigenous isolates of EPF in fruit fly *B. zonata* and its impact on adult mortality. This research provides a basis for implementing EPF against fruit fly, leading to dire need for eco-friendly pest management strategies.

Methods

Acquisition of fruit fly population

Number of larvae, pupa, and adult of *B. zonata* was acquired from Insect Chemical Ecology Laboratory, Department of Entomology, UAF. The population of fruit flies was maintained on natural and artificial diet in the laboratory.

Fungal isolation and purification

EPF were isolated from soil samples obtained from different guava growing areas of Pakistan with modified insect bait technique (Nishi et al. 2017). In this method, fungi-infected cadavers with powdery growth were collected from different guava orchards. Mycelia from infected larvae were transferred to Sabraud Dextrose Agar Yeast (SDAY) medium with the help of inoculating needle and kept in incubator at $26 \pm 1^\circ\text{C}$ and 70% RH. Fungi were further purified using the hyphal tip technique, as described by Brown (1924).

Fungal identification

The morphological identification of obtained EPF isolates was done on SDAY medium incubated at $26 \pm 1^\circ\text{C}$ with 70% RH after 7–10 days. Morphological features, i.e. colony color, colony size, colony texture, spore shape and size were examined using a compound microscope (Meiji Techno) fitted with a digital camera, in accordance with the identification guidelines outlined by Humber (1997).

Fungal culture preparations

A seven-day-old fungal culture grown on SDAY was inoculated to Sabouraud Dextrose Broth (SDB) and incubated at $26 \pm 1^\circ\text{C}$ with 60% relative humidity on a rotary shaker for 14 days. The resulting mycelial mat and conidia were homogenized, filtered through muslin cloth, and mixed with sterile distilled water containing 0.01% Tween 80. Conidial concentration was measured using a hemocytometer under a Meiji Techno microscope. The final stock suspension was adjusted to 1×10^9 conidia/mL and stored at 4°C until further use.

EPF screening bioassay against larva, pupa and adult

The screening bioassay of entomopathogenic fungi (EPF) was conducted to evaluate their pathogenicity against larval, pupal, and adult stages of fruit flies using the same EPF isolates. For the larval and pupal bioassays, experiments were performed in plastic boxes containing 25 g of sterilized soil. A water suspension containing 1×10^9 viable conidia per mL was prepared for each EPF isolate, and 1 mL of this suspension was poured and thoroughly mixed into the soil of each treatment box. In the control treatments, 2 mL of Silwet L-77 solution was used instead of the fungal suspension, and soil moisture was maintained at 10%. For larval assays, ten third-instar larvae were released into each box containing EPF-treated soil, while for pupal assays, ten pupae were similarly introduced into the treated soil. All boxes were covered and incubated at 25–27°C. Larval and pupal mortality were assessed indirectly by counting the number of adults that emerged and subtracting them from the initial number of insects exposed. Each treatment represented a single EPF isolate and was replicated three times, and the entire experiment was repeated twice to ensure consistency.

For the adult bioassay, the same EPF isolates were screened against adult fruit flies. The bioassay was performed in sterilized Petri plates, into which 1 mL of the fungal suspension containing 1×10^9 viable conidia per mL was poured and evenly distributed by shaking. Plates were allowed to dry until the solution desiccated. For the control group, only water was used in the Petri plates. Adult flies aged 1 to 4 days were immobilized by chilling them at 4°C, after which 3 µL of the fungal suspension was applied to the dorsal thorax of each fly using a micropipette. Treated flies were then transferred to cages containing artificial food and allowed to recover under controlled conditions of $27 \pm 2^\circ\text{C}$ temperature, $75 \pm 5\%$ relative humidity, and a 12-hour light, 12-hour dark photo period. Adult survival was monitored daily for 14 days by counting the number of live flies in each setup. Each treatment in the adult bioassay represented a single EPF isolate and was replicated three times, with the entire experiment repeated twice for validation.

Bioassay for different doses against different life stages of *B. zonata*

The EPF showing significant mortality were selected and evaluated at different concentrations (1×10^5 , 1×10^6 , 1×10^7 and 1×10^8 conidia/mL) for larval, pupal and adult stages. The treatment application was the same as in previous bioassays. Each fungal isolate was tested at various concentrations, with each concentration serving as a single treatment. Experiment was repeated thrice, with a total of 60 insects i.e. 20 in each replication. Larval mortality was evaluated based on adult emergence, following the previously described method. The bioassay concluded on the 4th day after the emergence of adults in the control group. Mortality in adults was documented at intervals of 2, 4, 6, 8, 10, 12, and 14-day post-treatment.

Horizontal transmission bioassay

The ability of the isolates to transmit horizontally was assessed using adult specimens of *Bactrocera zonata*. Adult flies were collected and separated by sex. Each fungal isolate was tested through four different inoculation combinations. The experimental setup was comprised of Petri dishes (9 cm

diameter × 1.5 cm depth) coated with 1 mL of each entomopathogenic fungus (EPF) suspension at a concentration of 1×10^8 conidia mL⁻¹. The coated Petri plates were kept on a rotary shaker to dry fungal suspension completely. Insects either male or female were exposed to the conidial suspension, while non-inoculated insects were treated with sterilized distilled water. To observe conidial transmission between adult flies, four groups insect combinations were made. In first group both inoculated male and female were kept. While in second group, 20 ten-day-old males were inoculated. Twenty-four hours after inoculation, these infected males were introduced to 20 non-inoculated females (seven-day-old) and kept in cages for potential transmission. In third group, analogous set of experiments was conducted using inoculated female flies and non- inoculated males. Additionally, a fourth group consisting of 20 fungus-free males and females was maintained under identical conditions. The experiments were conducted under controlled conditions at 25°C, 50–60% relative humidity, and a 14:10-hour light-dark photoperiod. Mortality assessments were performed at every 2 day interval until 14-days post-treatment. To prevent cross-contamination, dead adults were removed from cages daily.

Statistical analysis

Mortality for the treated group was adjusted using the Abbott formula (Abbott, 1925). For analysis of mortality data ANOVA was performed. Subsequently, the Tukey HSD (multiple comparison test) was employed as a post hoc analysis to find significant difference between the means.

Results

Morphological identification of the fungus

On the basis of morphological characters, the fungus isolated from diseased cadavers of fruit fly was identified as *Beauveria bassiana*. On PDA medium, the fungal colonies exhibited a white front, a yellow reverse, and a smooth, powdery to cottony texture with raised elevation. Conidia were round to spherical in shape, hyaline, smooth- walled, with a septate hyphae and branched conidiophores.

Effect of different isolates of *B. basiana* against 3rd Instar larvae, pupa and adults:

Against larva: Generally, FMB-BB-FF 04 of *B. basiana* showed the highest mortality (70 ± 1.82). while FMB-BB-FF 02 and FMB-BB-FF 03 showed $68 \pm 1.82A$ and $60 \pm 1.82B$ mortality respectively. While minimum mortality ($55 \pm 1.82B$) was shown by FMB-BB-FF 01. There was no significant difference among FMB-BB-FF 04 and FMB-BB-FF 02 and FMB-BB-FF 01 and FMB-BB-FF 03 (Fig:1). **Against pupa:** Generally, FMB-BB-FF 04 of *B. basiana* showed the highest mortality ($77 \pm 2.95A$). while FMB-BB-FF 02 and FMB-BB-FF 03 showed $73 \pm 2.95AB$ and $64 \pm 2.95BC$ mortality respectively. While minimum mortality ($58 \pm 2.95C$) was shown FMB-BB-FF 01. All isolates were significantly different from each other (Fig:1). **Against adult:** Generally, FMB-BB-FF 04 of *B. basiana* showed the highest mortality ($92 \pm 2.23A$). while FMB-BB-FF 02 and FMB-BB-FF 03 showed $88 \pm 2.23A$ and $76 \pm 2.23B$ mortality respectively. While minimum mortality ($70 \pm 2.23B$) was shown by FMB-BB-FF 01. There was no significant difference

among FMB-BB-FF 04 and FMB-BB-FF 02 and FMB-BB-FF 01 and FMB-BB-FF 03 (Fig:1). All four isolates were chosen for follow-up bioassays.

Effect of different concentrations of *B. basiana* against 3rd Instar larvae, pupa and adults:

All four concentrations tested showed variable range of mortality, with mean percentage mortality from 55–94% for larvae, 53–92% for pupa and from 56–96% for adults of *B. zonata*.

Against larva:

FMB-BB-FF 04 showed the highest mortality ($94 \pm 1.52A$) at concentration 1×10^8 spores/mL. While at concentration 1×10^6 spore/mL and concentration 1×10^7 spore/mL mortality range were $74 \pm 1.00A$ and $85 \pm 1.46A$. Least mortality ($70 \pm 1.32 A$) was shown at concentration 1×10^5 spores/mL. FMB-BB-FF 02 showed the highest mortality ($90 \pm 1.52AB$) at concentration 1×10^8 spores/mL. While at concentration 1×10^6 spore/mL and concentration 1×10^7 spore/mL mortality range were $71 \pm 1.00A$ and $84 \pm 1.46A$. Least mortality ($65 \pm 1.32 AB$) was shown at concentration 1×10^5 spores/mL. FMB-BB-FF 03 showed the highest mortality ($85 \pm 1.52BC$) at concentration 1×10^8 spore/mL. While at concentration 1×10^6 spores/mL and concentration 1×10^7 spores/mL mortality range were $60 \pm 1.00B$ and $75 \pm 1.46B$. Least mortality ($60 \pm 1.32 BC$) was shown at concentration 1×10^5 spore/mL. FMB-BB-FF 01 showed the highest mortality ($81 \pm 1.52C$) at concentration 1×10^8 spore/mL. While at concentration 1×10^6 spores/mL and concentration 1×10^7 spores/mL mortality range were $59 \pm 1.00B$ and $71 \pm 1.46B$. Least mortality ($55 \pm 1.32 C$) was shown at concentration 1×10^5 spores/mL (Fig:2).

Against pupa:

FMB-BB-FF 04 showed the highest mortality ($92 \pm 1.07A$) at concentration 1×10^8 . While at concentration 1×10^6 spore/mL and concentration 1×10^7 spore/mL mortality range were $70 \pm 1.44A$ and $84 \pm 2.37A$. Least mortality ($67 \pm 1.40A$) was shown at concentration 1×10^5 spore/mL. FMB-BB-FF 02 of *B. basiana* showed the highest mortality ($89 \pm 1.07A$) at concentration 1×10^8 spore/mL. While at concentration 1×10^6 spore/mL and concentration 1×10^7 spore/mL mortality range were $69 \pm 1.44A$ and $82 \pm 2.37AB$. Least mortality ($63 \pm 1.40A$) was shown at concentration 1×10^5 spore/mL. FMB-BB-FF 03 of *B. basiana* showed the highest mortality ($85 \pm 1.07B$) at concentration 1×10^8 spore/mL. While at concentration 1×10^6 spore/mL and concentration 1×10^7 spore/mL mortality range were $58 \pm 1.44B$ and $75 \pm 2.37BC$. Least mortality ($58 \pm 1.40B$) was shown at concentration 1×10^5 spore/mL. FMB-BB-FF 01 showed the highest mortality ($78 \pm 1.07C$) at concentration 1×10^8 spore/mL. While at concentration 1×10^6 spore/mL and concentration 1×10^7 spore/mL mortality range were $58 \pm 1.44B$ and $69 \pm 2.37C$. Least mortality ($53 \pm 1.40B$) was shown at concentration 1×10^5 spore/mL (Fig:3).

Against adult:

FMB-BB-FF 04 showed the highest mortality ($96 \pm 1.23A$) at concentration 1×10^8 spore/mL. While at concentration 1×10^6 spore/mL and concentration 1×10^7 spore/mL mortality range were $75 \pm 1.47A$ and $86 \pm 2.19A$. Least mortality ($71 \pm 1.59A$) was shown at concentration 1×10^5 spore/mL. FMB-BB-FF 02 showed the highest mortality ($92 \pm 1.23A$) at concentration 1×10^8 spore/mL. While at concentration 1×10^6 spore/mL and concentration 1×10^7 spore/mL mortality range were $72 \pm 1.47A$ and $85 \pm 2.19A$. Least mortality ($67 \pm 1.59A$) was shown at concentration 1×10^5 spore/mL. FMB-BB-FF 03 showed the highest mortality ($87 \pm 1.23B$) at concentration 1×10^8 spore/mL. While at concentration 1×10^6 spore/mL and concentration 1×10^7 spore/mL mortality range were $62 \pm 1.47B$ and $76 \pm 2.19B$. Least mortality ($61 \pm 1.59B$) was shown at concentration 1×10^5 spore/mL. FMB-BB-FF 01 showed the highest mortality ($84 \pm 1.23B$) at concentration 1×10^8 spore/mL. While at concentration 1×10^6 spore/mL and concentration 1×10^7 spore/mL mortality range were $60 \pm 1.47B$ and $73 \pm 2.19B$. Least mortality ($56 \pm 1.59B$) was shown at concentration 1×10^5 spore/mL (Fig:4).

Horizontal transmission bioassay:

There was Significant differences in male and female mortality for all tested isolates. The highest mortality rates were documented when both sexes were inoculated. However, this treatment did not show a significant difference compared to the treatment non-inoculated females with inoculated males. The spore transmission from inoculated male to non-inoculated female was more obvious compared to the transmission from inoculated females to non-inoculated males (Fig:5).

Discussion

Isolating entomopathogens from naturally infected fruit flies and utilizing these native microorganisms as biopesticides can effectively suppress pest populations (Eski and Gezgin 2022). However, commercially available entomopathogenic fungi (EPFs) often face limitations in adaptability across diverse agroecological zones. This challenge can be addressed by employing native isolates with high virulence and environmental compatibility. In the current study, the entomopathogenic fungus isolated from fruit fly cadavers was identified as *Beauveria bassiana* based on both morphological traits. Our study specifically targeted the 3rd instar larvae of *Bactrocera zonata*, as the earlier instars reside inside the fruit, while the third instars exit the fruit and fall into the soil to pupate (Cossentine et al. 2010). Consequently, targeting this final larval stage represents an effective control strategy, often leading to elevated levels of pupal mortality. The present study confirmed the pathogenic efficacy of native *Beauveria bassiana* isolates against various developmental stages of the fruit fly (*Bactrocera zonata*) under laboratory conditions. Previous investigations have explored the susceptibility of *B. zonata* life stages to *B. bassiana* under both laboratory (Ahmed et al. 2022) and field conditions (Khan et al. 2017). Hussein et al. (2018) concluded that adult *B. zonata* were more vulnerable to *M. anisopliae* and *B. bassiana* than larvae or pupae. Susceptibility differences across EPF strains and insect stages are linked to factors like conidial attachment, cuticle structure, and immune evasion (Ekesi 2016). Lower mortality in larvae and older pupae may be due to thicker, more sclerotized cuticles. Despite this, high adult

mortality was observed after emergence from treated pupae, aligning with results in *C. capitata* and *A. ludens* (Quesada-Moraga et al. 2006).

There was no significant difference in pathogenicity between two *Beauveria bassiana* isolates (FMB-BB-FF 01 and FMB-BB-FF 03), likely due to similar levels of enzyme and toxin production. EPFs produce enzymes and secondary metabolites, and strains with higher enzyme activity tend to be more virulent (Prasetia et al. 2023). Variations in virulence are often linked to differences in the production of toxins like beauvericin, enniatin, oosporein, and bassianolide (Deka et al. 2021), with beauvericin being the primary toxin produced by *B. bassiana* (Safavi 2013). This study further demonstrated that indigenous *B. bassiana* isolates induced cumulative mortality of up to 98% in larvae, pupae, and adults within 14 days at a concentration of 1×10^8 conidia/mL, with mortality increasing in correlation with higher conidial concentrations and extended exposure times. Consistent with the findings of Yasin et al. (2019), the mortality of *Bactrocera zonata* larvae increased with higher concentrations of conidia. Similarly, the *B. zonata* showed varying degrees of susceptibility to different *Beauveria bassiana* isolates, as reported in the same study. For effective pest management, however, it is essential to select isolates that possess both high virulence and adaptability to local environmental conditions. Therefore, the application of microbial bioagents, particularly *B. bassiana*, provides a basis for sustainable and long-lasting control of fruit fly populations.

An entomopathogenic fungi isolate with efficient dispersal and the ability to induce secondary infections that result in mortality serve as an effective biocontrol agent. Horizontal transmission can occur through physical contact or mating interactions. In fruit flies, grooming behavior following exposure to entomopathogenic fungi significantly influences the retention of conidia, which in turn affects the success of horizontal transmission. In the present study, the research evaluated the potential for horizontal transmission of these fungal isolates. *Bactrocera zonata* individuals treated with entomopathogenic fungi (donors) successfully transmitted the fungal conidia to untreated individuals (recipients), resulting in increased mortality rate. Similar findings have been reported in other species: Quesada-Moraga et al. (2006) demonstrated effective horizontal transmission in populations of *Ceratitis capitata* in vitro conditions. Moreover, effective horizontal transmission of *Beauveria bassiana* has also been documented in other insect species such as the kissing bug *Triatoma infestans* and the Mexican fruit fly *Anastrepha ludens* (Goble et al. 2011). Ongoing research to identify highly virulent local *Beauveria bassiana* isolates and develop effective field application methods is essential for enhancing the efficiency and cost-effectiveness of biopesticides. Selecting superior strains is crucial for success (Camara et al. 2022). Further evaluation of the four selected *B. bassiana* strains is needed, with potential for safe and eco-friendly pest control (Trizelia and Syahrawati 2023). Key advantages include broad pest targeting, safety for natural enemies, compatibility with other methods, and lower environmental impact compared to chemical insecticides.

Conclusion

The results of this study confirmed the pathogenicity of the indigenous *Beauveria bassiana* isolate against *Bactrocera zonata*. Under laboratory conditions, *B. bassiana* effectively reduced fruit fly populations, offering a promising, environmentally friendly alternative to synthetic insecticides and harmful chemical pesticides in current management programs.

Abbreviations

EPF Entomopathogenic fungi Bb- *Beauveria bassiana* FMB fungal molecular biology

Declarations

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Author Contribution

SR; conducted the experiments and wrote the manuscript; IUH supervised the research RAA; Analyzed the data; SM; provided the fruit flies used in the research M: Provided the technical guidance; FJ—Revised the manuscript. All authors read and approved the final manuscript.

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Figures

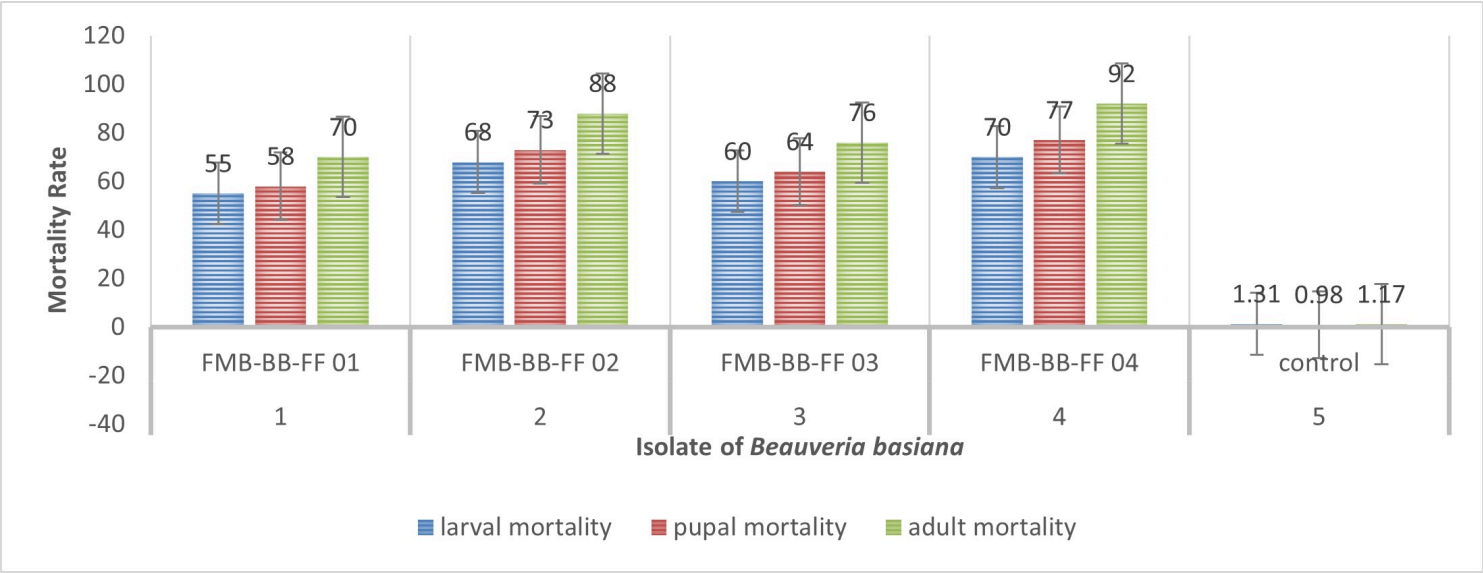


Figure 1

Percent larval, pupal and adult mortality (mean) of *B. zonata* when four different isolates of *B. bassiana* were applied.

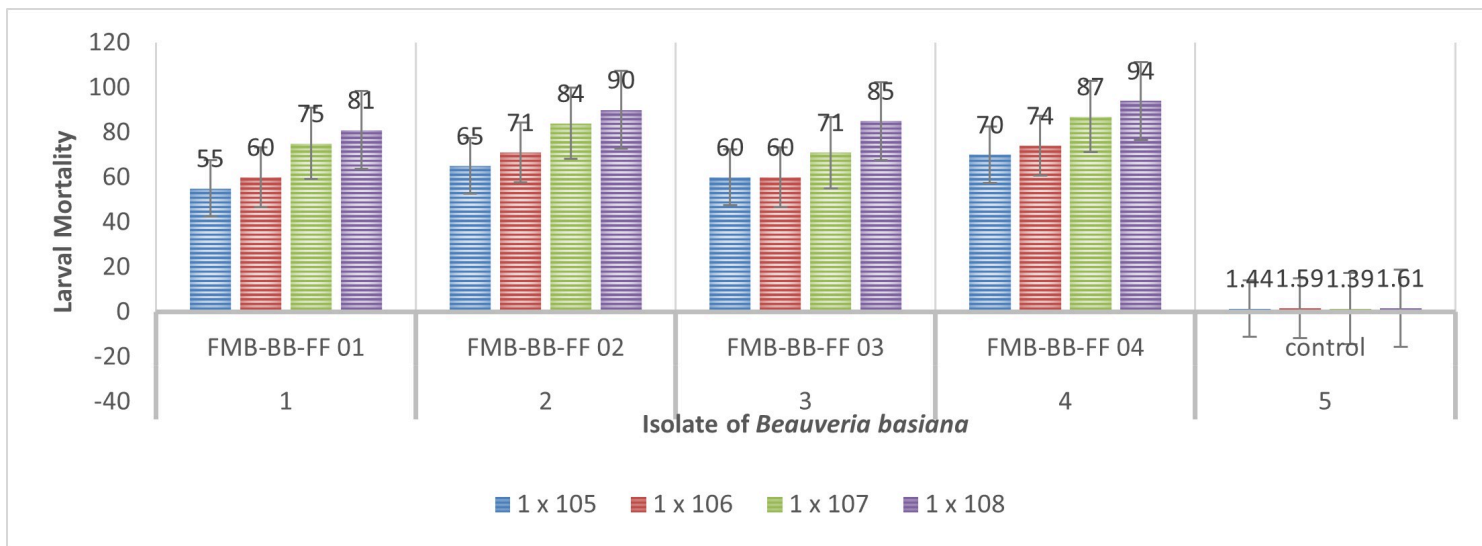


Figure 2

Percent larvae mortality (mean) of *B. zonata* when four different concentrations of four different isolates of *B. basiana* were applied.

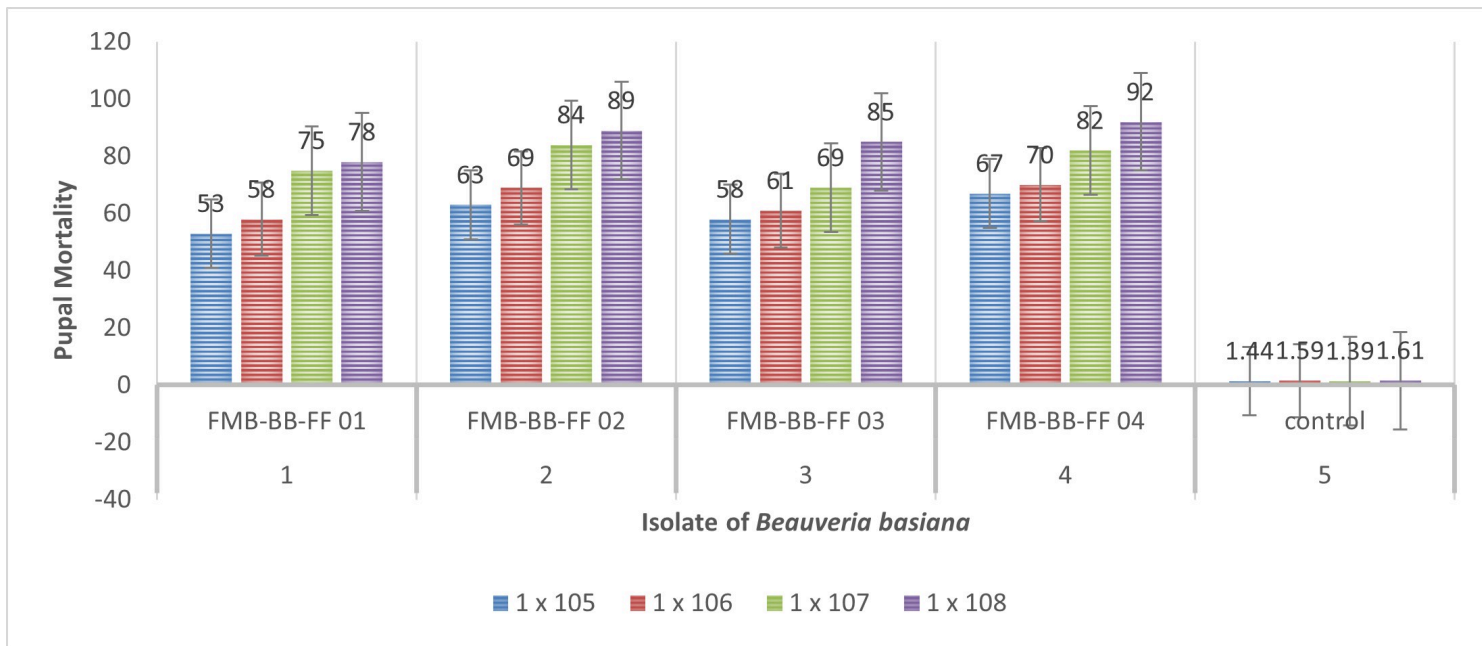


Figure 3

Percent pupal mortality (mean) of *B. zonata* when four different concentrations of four different isolates of *B. basiana* were applied.

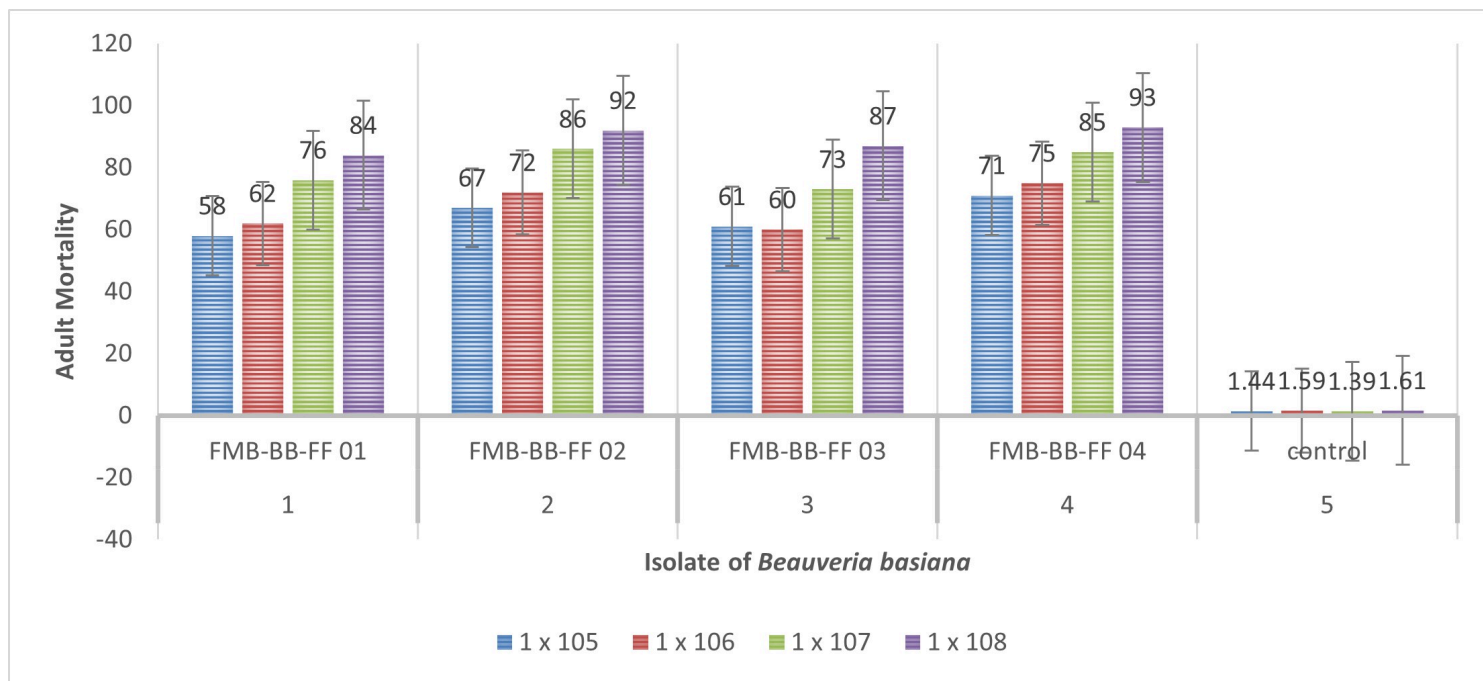


Figure 4

Percent adult mortality (mean) of *B. zonata* when four different concentrations of four different isolates of *B. basiana* were applied.

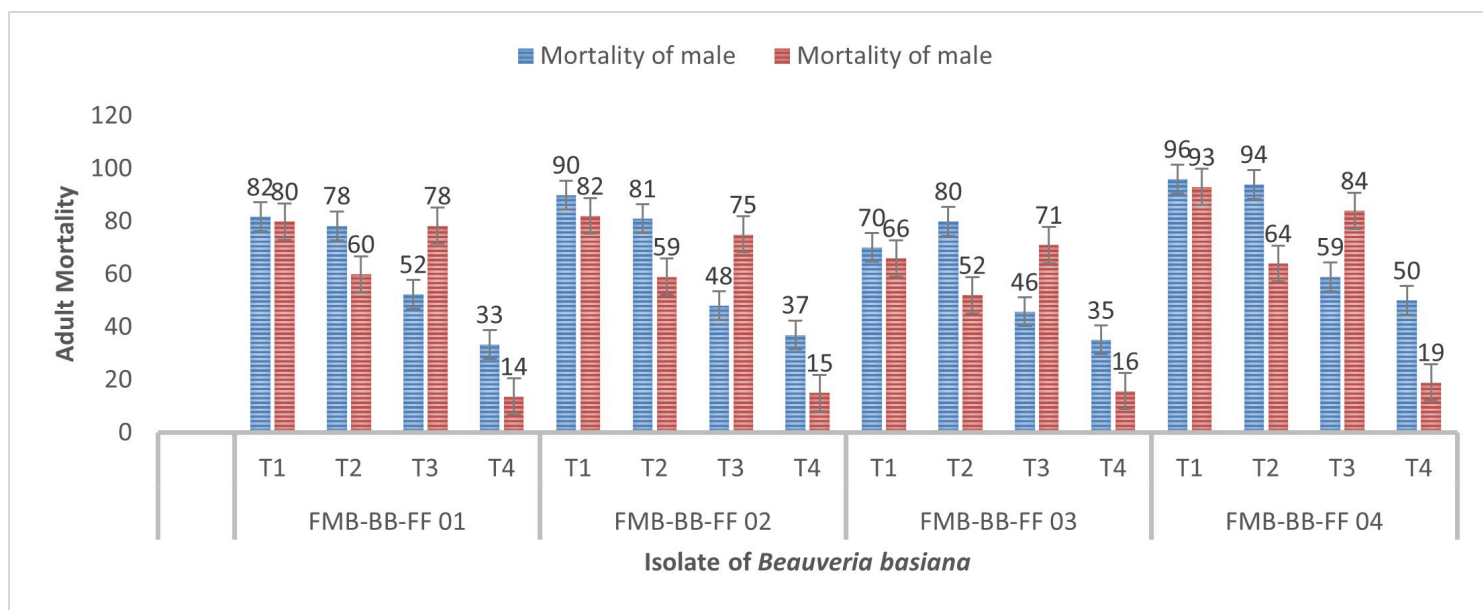


Figure 5

Percent mortality in adults of *B. zonata* caused by horizontal transmission of four isolates of *B. basiana*.